

# Xiaochuan Li

M.Sc. Human and AI · University of Technology Nuremberg

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## RESEARCH INTERESTS

Formal and experimental semantics; psycholinguistics; language processing and learning; computational semantics; language-model representation, evaluation, and intervention. Current case study: epistemic modality and discourse alternatives. Exploratory interest: selective property inheritance in noun modification.

## EDUCATION

**M.Sc., Human and Artificial Intelligence** 2025 – 2027

University of Technology Nuremberg (UTN)

Expected graduation: Summer 2027. Coursework: cognitive psychology, experimental design and statistics, interpretability, deep learning, epistemology.

**Student · Preparatory Studies** 2024 – 2025

University of Marburg · enrolled student; intensive German for graduate study

Comparing German with Chinese and English made grammatical differences—case, word order, and clause structure—concrete rather than abstract.

**B.A., Philosophy** 2020 – 2024

Shenzhen University

Senior thesis on Leibniz's *characteristica universalis* (advisor: Zang Yong). Coursework in philosophy of language, analytic philosophy, logic, and mathematical logic.

## MANUSCRIPTS AND RESEARCH PROJECTS

**Three Ways of Leaving  $p$  Unsettled** — analysis in development

A formal semantic and pragmatic comparison of might  $p$ , might  $p$  or might not  $p$ , and I don't know whether  $p$ , examining the commitments they create and the responses they license. Independent judgments on the diagnostic materials are pending.

**M.Sc. thesis: Epistemic possibilities in use** — scope under discussion

A proposed adult behavioural study would test whether the linguistic contrasts affect later access to a specific possibility and its updating when new evidence arrives. The primary focus and any computational component remain under discussion; data collection has not begun.

**Cross-Lingual Prototypicality Bias in Multimodal Evaluation Metrics and VLM Judges** — first-author; in revision for arXiv

Controlled cross-lingual evaluation across seven languages and two VLM families; about 12,600 judgments, item-clustered analysis, a human control, and public code. [GitHub](#) ↗

## RESEARCH EXPERIENCE

**Latent Control States** — ongoing collaboration with DFKI

Designed controlled dilemma comparisons separating outcome sensitivity, framing, and action preferences. Behavioural pilots are in place; mechanistic analysis is in progress.

## EXPLORATORY WORK

**Selective property inheritance.** When does a modifier cancel one default property while preserving others? Research framing in progress; no empirical results.

## METHODS AND TECHNICAL SKILLS

**Programming and data.** Python, NumPy, pandas, scikit-learn, Git, LaTeX.

**ML / NLP.** PyTorch, Hugging Face Transformers, model inference, dataset pipelines, LoRA fine-tuning.

**Interpretability.** Linear and structural probing, activation patching, steering, ablation (working proficiency).

**Experimental design and statistics.** Controlled stimuli, minimal pairs, preregistration, mixed-effects analysis (working proficiency).

## LANGUAGES

Mandarin Chinese (native), German (C1), English (C1), Japanese (learning).